Smit Chaudhary

Software Engineer · Quantum Algorithms Developer · Quantum Physicist

Rotterdamm The Netherlands

(+31) 06 1378 2478 smitchaudhary10@gmail.com https://smitchaudhary.github.io/ smitchaudhary 0000-0003-0243-6513

Experience

Quantum Algorithms Developer Amsterdam, The Netherlands

Pasqal 2022 - Present

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Quantum Computing Intern California, USA

Menten AI May 2022 - Sept 2022

- Developed a novel fully quantum generative adversarial network (QGAN) architecture for discrete and binary data generation
- Investigated and implemented quantum circuit features such as noise reuploading in the generator and auxiliary qubits in the discriminator to enhance expressivity
- Demonstrated the model's performance on synthetic datasets and low-energy states of Ising models, showing successful data reproduction and generalization capabilities

Education

Technische Universiteit Delft Delft, The Netherlands

Master of Science in Applied Physics August 2020 - July 2022

- **Thesis:** "Quantum Walk Based Qubit Mapping" (Supervisor: Prof. Sebastian Feld)
- **Relevant Coursework:** Applied Quantum Algorithms, Quantum Information, Quantum Computing Architecture
- **Research Focus:** Near term and variational quantum algorithms

Indian Institute of Technology, Kanpur Kanpur, India

Bachelor of Science in Physics July 2016 - May 2020

- **Relevant Coursework:** Quantum Computing, Quantum Field Theory, Statistical Mechanics, Probability and Statistics, Optics
- **Undergraduate Project:** "Bohmian Mechanics and Quantum Information" (Supervisor: Prof. Kaushik Bhattacharya)
- **Research Experience:** Summer Research at IISER Kolkata under Prof. Prasanta Panigrahi

Selected Publications

Solving Fluid Dynamics Equations with Differentiable Quantum Circuits

S. Chaudhary, G. T. Balducci, O. Kyriienko, P. K. Barkoutsos, L. Cardarelli, A. A. Gentile

Proceedings of the 35th Parallel CFD International Conference 2024, May 2025

Quantum Circuit Training with Growth-Based Architectures

C. Duffy, **S. Chaudhary**, G. V. Velikova

arXiv preprint, Nov 2024

Towards a scalable discrete quantum generative adversarial neural network

S. Chaudhary, P. Huembeli, I. MacCormack, T. L. Patti, J. Kossaifi, A. Galda

arXiv preprint, 2022

Skills

Programming	**Python**, **C/C++**, Rust, Verilog
Languages	**English**, Hindi, Gujarati
Libraries & Utilities	**Git**, **PyTorch**, Pennylane, Tensorflow, Jax, Cirq, Qiskit